



2006

YEAR 11 EXAMINATION

Preliminary Mathematics

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using black or blue pen
- Board approved calculators may be used
- All necessary working should be shown in every question

Total Marks – 84

Attempt Questions 1–7

All questions are of equal value

NAME: _____ TEACHER: _____

STUDENT NUMBER: _____

QUESTION	MARK
1	/12
2	/12
3	/12
4	/12
5	/12
6	/12
7	/12
TOTAL	/84

Total Marks – 84

Attempt Questions 1–7

All questions are of equal value

Begin each question on a NEW PAGE.

Question 1: (12 marks)

Marks

- a) Find the value of $\frac{4 \cdot 23}{\sqrt{6 \cdot 14 - 1 \cdot 78}}$ correct to 3 significant figures. **2**
- b) Express $0.\dot{2}\dot{1}$ as a fraction in its simplest form. **1**
- c) Show that $\frac{1}{3-\sqrt{2}} + \frac{1}{3+\sqrt{2}}$ is a rational number. **2**
- d) Simplify $\frac{2}{3} - \frac{x-1}{4}$ **2**
- e) Solve the simultaneous equations:
$$\begin{aligned} x - y &= 2 \\ 3x + 2y &= 1 \end{aligned}$$
 2
- f) Evaluate $\left(\frac{1}{16}\right)^{-\frac{3}{4}}$ **1**
- g) Factorise $125a - ab^3$ **2**

Question 2: (12 marks) Start a new page

Marks

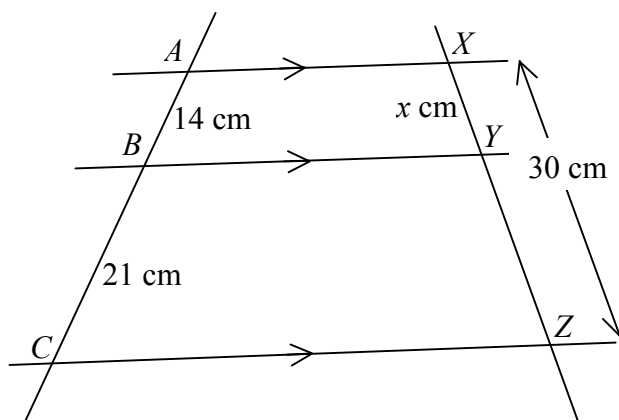
- a) Simplify the expression $\frac{6}{x^2 + 6x + 5} - \frac{4}{x^2 + 5x}$ **3**
- b) Solve the inequality $|16 - 4x| > 8$ and graph your solution on the number line. **3**
- c) Solve $(2x + 3)^2 = 25$ **2**
- d) Factorise completely $x^4 - x^2 - 12$ **2**
- e) Find the coordinates of the centre of the circle $x^2 + y^2 - 2x + 6y + 1 = 0$. **2**

Question 3: (12 marks) Start a new page

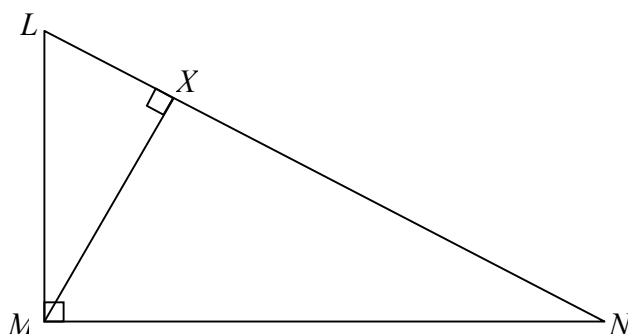
Marks

- a) In the diagram, $AX \parallel BY \parallel CZ$.
 $AB = 14$ cm; $BC = 21$ cm; $XZ = 30$ cm and $XY = x$ cm.
 Find the length of XY giving reasons.

2



b)



$\triangle LMN$ is right angled at M . X lies on LN such that $MX \perp LN$.

i) Prove that $\triangle LMX$ is similar to $\triangle LNM$.

2

ii) Hence prove that $LM^2 = NL \times LX$.

2

c) Sketch each of the following, showing their important features.

i) $y = \sqrt{25 - x^2}$

2

ii) $y = |2x - 1|$

2

iii) $y = x(x + 2)(1 - x)$

2

Question 4: (12 marks) Start a new page**Marks**

- a) A piecemeal function is defined as

$$f(x) = \begin{cases} -x^2, & x < 0 \\ 5x - 4, & x \geq 0 \end{cases}$$

Evaluate $f(-2) + f(0) - f(1)$

2

- b) State the domain of the function $y = \sqrt{2x + 5}$.

1

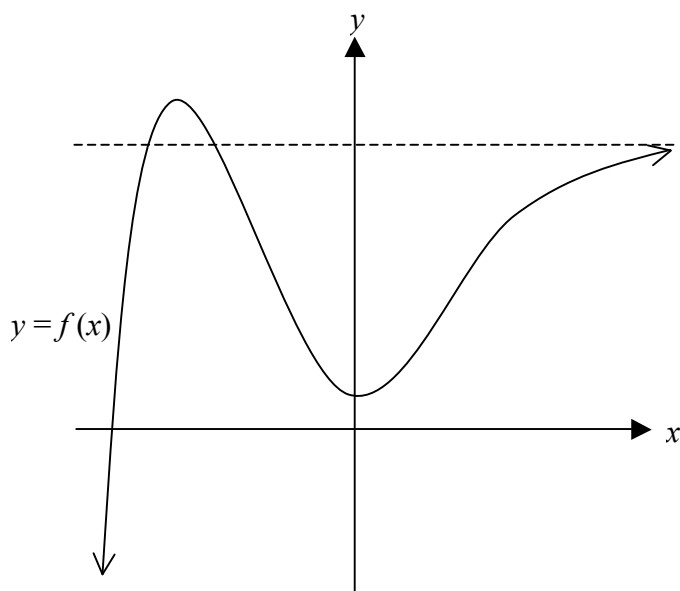
- c) Sketch $y = 5^x - 3$ and hence state the range of the function.

3

- d) Sketch the region in the number plane which $y \leq 4 - x^2$ and $y > -2$ hold simultaneously.

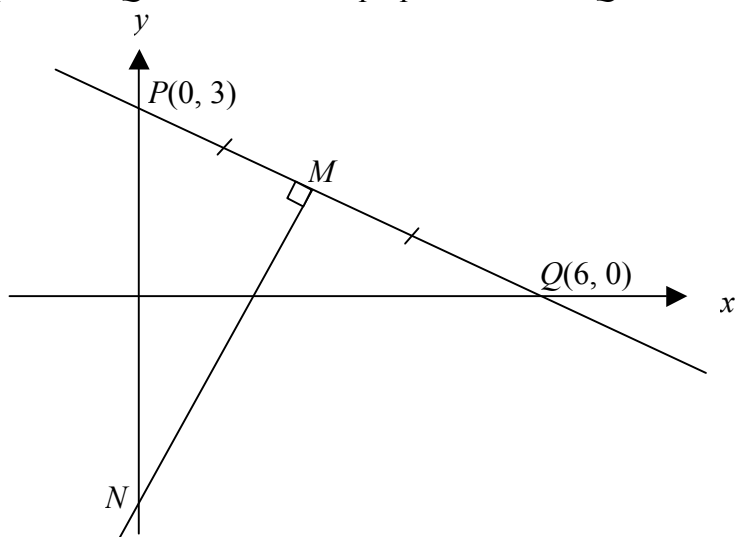
3

- e) The diagram illustrates the graph of $y = f(x)$. Copy or trace the graph onto your examination page and, on a new set of axes directly below, sketch the graph of its gradient function, $y = f'(x)$.

3

Question 5: (12 marks) Start a new page**Marks**

- a) The diagram shows the points $P(0, 3)$ and $Q(6, 0)$. The point M is the midpoint of PQ . The line MN is perpendicular to PQ and meets the y -axis at N .



- | | | |
|------|--|---|
| i) | Show that the gradient of PQ is $-\frac{1}{2}$. | 1 |
| ii) | Find the coordinates of M . | 2 |
| iii) | Find the equation of MN . | 2 |
| iv) | Show that the coordinates of N are $(0, -\frac{9}{2})$. | 1 |
| v) | The quadrilateral $PNQR$ is a parallelogram. Find the coordinates of R . | 1 |
- b) Find the perpendicular distance from the point $A(3, -2)$ to the line $y = 2x + 5$. 2
- c) Find the equation of the line which passes through the point of intersection of the lines $2x + 3y = 5$ and $3x - 4y = 7$ and which also passes through the point $(-1, 3)$. 3

Question 6: (12 marks) Start a new page**Marks**

- | | | |
|-----|--|---|
| a) | Find the value of θ if $0^\circ \leq \theta \leq 90^\circ$ and $\tan 20^\circ = \cot(\theta + 30^\circ)$ | 1 |
| b) | If $\sin x^\circ = \frac{4}{5}$ and $\cos x^\circ < 0$, find the value of $\tan x^\circ$. | 2 |
| c) | Prove that $\cot^2 \theta (1 - \cos^2 \theta) = \cos^2 \theta$ | 3 |
| d) | If $0^\circ \leq \theta \leq 360^\circ$, find all the solutions for θ for $2 \sin \theta \cos \theta - \sin \theta = 0$. | 3 |
| e) | A boat leaves port P and steams 80 nautical miles on a bearing of 110° . It then turns to a bearing of 190° and steams a further 100 nautical miles. | |
| i) | Draw a diagram illustrating this information. | 1 |
| ii) | How far is the boat from P to the nearest nautical mile? | 2 |

Question 7: (12 marks) Start a new page**Marks**

a) Differentiate the following:

i) $y = 5x^2 - 3x$

1

ii) $f(x) = \frac{(x^2 + 3)(5x - 1)}{x^2}$

3b) Consider the curve $y = x + \frac{8}{x}$.i) Find the gradient of the curve at the point P where $x = 2$.**2**ii) What are the co-ordinates of P .**1**iii) Find the equation of the normal to the curve at the point P .**2**c) At what point on the curve $f(x) = \frac{x^2}{4} - 5$ is the tangent parallel to $3x - y + 2 = 0$?**3****End of paper**

Solutions

Question one

$$a) 2.02580... \\ \approx 2.03$$

$$b) \frac{21}{99} \\ = \frac{7}{33}$$

$$c) \frac{1}{3-\sqrt{2}} + \frac{1}{3+\sqrt{2}} \\ = \frac{3+\sqrt{2} + 3-\sqrt{2}}{(3-\sqrt{2})(3+\sqrt{2})} \\ = \frac{6}{9-2} \\ = \frac{6}{7}$$

which is rational

$$d) \frac{2}{3} - \frac{x-1}{4} \\ = \frac{2(4) - 3(x-1)}{12} \\ = \frac{8-3x+3}{12} \\ = \frac{11-3x}{12}$$

$$e) \begin{aligned} 2x-2y &= 4 \\ 3x+2y &= 1 \\ 5x &= 5 \\ x &= 1 \end{aligned}$$

$$\begin{aligned} x=1, 2-2y &= 4 \\ 2y &= -2 \\ y &= -1 \end{aligned}$$

$$\begin{cases} x=1 \\ y=-1 \end{cases}$$

$$f) (16)^{\frac{3}{4}} \\ = 8$$

$$g) a(125-b^3) \\ = a(5-b)(25+5b+b^2)$$

Question two

$$a) \frac{6}{(x+5)(x+1)} - \frac{4}{x(x+5)} \\ = \frac{6(x) - 4(x+1)}{x(x+5)(x+1)} \\ = \frac{6x - 4x - 4}{x(x+5)(x+1)} \\ = \frac{2x-4}{x(x+5)(x+1)}$$

$$b) |16-4x| > 8 \\ 16-4x > 8 \quad \text{or} \quad 16-4x < -8 \\ -4x > -8 \quad \quad \quad -4x < -24 \\ x < 2 \quad \quad \quad x > 6$$



$$c) \begin{aligned} (2x+3)^2 &= 25 \\ 2x+3 &= \pm 5 \\ 2x+3 &= 5 & 2x+3 &= -5 \\ 2x &= 2 & 2x &= -8 \\ x &= 1 & x &= -4 \end{aligned}$$

$$x = 1, -4$$

Question two

d) let $m = x^2$

$$m^2 - m - 12$$

$$(m-4)(m+3)$$

$$\therefore (x^2-4)(x^2+3)$$

$$= (x-2)(x+2)(x^2+3)$$

e) $x^2 - 2x + 1 + y^2 + 6y + 9 = -1 + 1 + 9$

$$(x-1)^2 + (y+3)^2 = 9$$

Centre $(1, -3)$

Question three

a) $\frac{AB}{BC} = \frac{XY}{YZ}$ a family of parallel lines will cut all transversals in the same ratio

$$\frac{14}{21} = \frac{x}{(30-x)}$$

$$\frac{2}{3} = \frac{x}{(30-x)}$$

$$60 - 2x = 3x$$

$$5x = 60$$

$$x = 12$$

$$XY = 12 \text{ cm}$$

b) In $\triangle LMX$, $\triangle LNM$

$$\angle MLX = \angle NLM \text{ Common}$$

$$\angle LXM = \angle LMN \text{ both } 90^\circ$$

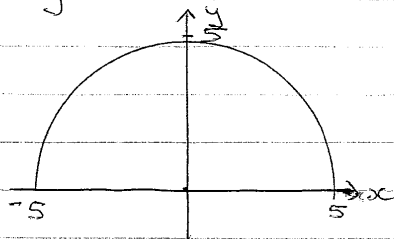
$\therefore \triangle LMX \parallel \triangle LNM$ equiangular

$$\frac{LM}{LN} = \frac{LX}{LM} \text{ ratio of sides of similar triangles}$$

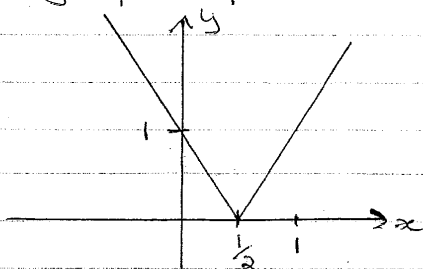
$$\therefore LM^2 = NL \cdot LX$$

Question three

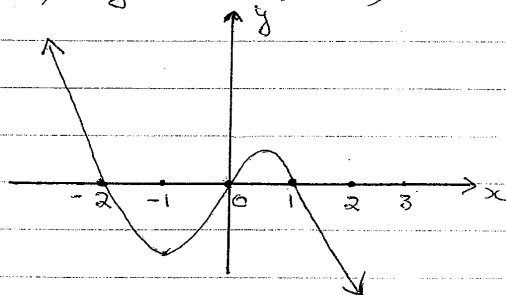
c) i) $y = \sqrt{25 - x^2}$



(ii) $y = |2x - 1|$



(iii) $y = x(x+2)(1-x)$

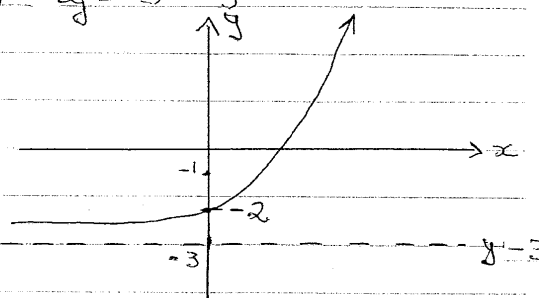


Question four

a) $f(-2) + f(0) - f(1)$
 $= -4 + -4 - 1$
 $= -9$

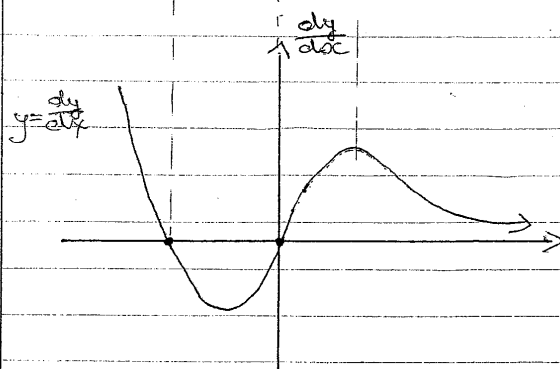
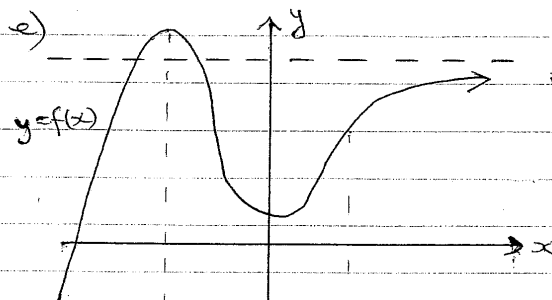
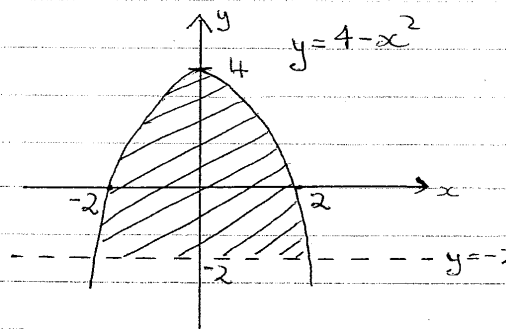
b) domain $\{x : x \geq -\frac{5}{2}\}$

c) $y = 5^x - 3$



range: $\{y : y > -3\}$

d) $y = 4 - x^2$



Question five

a) i) $m = \frac{0-3}{6-0}$
 $= \frac{-3}{6}$
 $= -\frac{1}{2}$

(ii) $M: \left(\frac{0+6}{2}, \frac{3+0}{2} \right)$

$$= \left(\frac{6}{2}, \frac{3}{2} \right)$$

$$= (3, 1\frac{1}{2})$$

(iii) gradient MN is 2

$$y - y_1 = m(x - x_1)$$

$$y - \frac{3}{2} = 2(x - 3)$$

$$2y - 3 = 4(x - 3)$$

$$2y - 3 = 4x - 12$$

$$0 = 4x - 2y - 9$$

(iv) $x=0, 4(0) - 2y - 9 = 0$

$$2y = -9$$

$$y = -\frac{9}{2}$$

$$\therefore N(0, -\frac{9}{2})$$

(v) $R(6, 7\frac{1}{2})$

b) $d = \left| \frac{ax + by + c}{\sqrt{a^2 + b^2}} \right|$

$$a = 2, b = -1, c = 5$$

$$d = \left| \frac{2(3) - 1(-2) + (5)}{\sqrt{(2)^2 + (-1)^2}} \right|$$

$$= \left| \frac{6 + 2 + 5}{\sqrt{4+1}} \right|$$

$$= \frac{13}{\sqrt{5}}$$

c) $(2x + 3y - 5) + k(3x - 4y - 7) = 0$

$$(2(-1) + 3(3) - 5) + k(3(-1) - 4(3) - 7) = 0$$

$$(-2 + 9 - 5) + k(-3 - 12 - 7) = 0$$

$$2 + k(-22) = 0$$

$$22k = 2$$

$$k = \frac{1}{11}$$

$$(2x + 3y - 5) + \frac{1}{11}(3x - 4y - 7) = 0$$

$$11(2x + 3y - 5) + (3x - 4y - 7) = 0$$

$$22x + 33y - 55 + 3x - 4y - 7 = 0$$

$$25x + 29y - 62 = 0$$

Question six

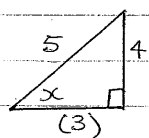
a) $\tan 20^\circ = \cot 70^\circ$

$$\therefore 70 = \theta + 30^\circ$$

$$\theta = 40^\circ$$

OR $20^\circ + 30^\circ + \theta = 90^\circ \therefore \theta = 40^\circ$

b) 2nd quadrant $\therefore \tan x < 0$



$$\tan x^\circ = -\frac{4}{3}$$

c) $\cot^2 \theta (1 - \cos^2 \theta) = \cos^2 \theta$

$$\text{LHS} = \frac{\cos^2 \theta}{\sin^2 \theta} \times (\sin^2 \theta)$$

$$= \cos^2 \theta$$

$$= \text{RHS}$$

Question six

d) $2 \sin \theta \cos \theta - \sin \theta = 0$

$\sin \theta (2 \cos \theta - 1) = 0$

$\sin \theta = 0$ or $2 \cos \theta - 1 = 0$

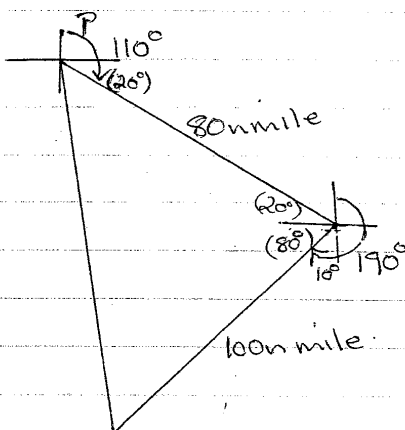
$\theta = 0^\circ, 180^\circ, 360^\circ$ $2 \cos \theta = 1$

$\cos \theta = \frac{1}{2}$

$\theta = 60^\circ, 300^\circ$

$\theta = 0^\circ, 60^\circ, 180^\circ, 300^\circ, 360^\circ$

e)



$d^2 = 80^2 + 100^2 - 2(80)(100) \cos 100^\circ$

$d = 138.4859 \dots$

the distance is approx

> 138 nautical miles.

Question seven

a) $y = 5x^2 - 3x$

$\frac{dy}{dx} = 10x - 3$

(ii) $f(x) = \frac{5x^3 - x^2 + 15x - 3}{x^2}$

$= 5x - 1 + 15x^{-1} - 3x^{-2}$

$f'(x) = 5 - 15x^{-2} + 6x^{-3}$

$= 5 - \frac{15}{x^2} + \frac{6}{x^3}$

b) $y = x + 8x^{-1}$

$\frac{dy}{dx} = 1 - 8x^{-2}$

$= 1 - \frac{8}{x^2}$

$x = 2, \frac{dy}{dx} = 1 - \frac{8}{4}$

$= -1$

the gradient at P is -1

(ii) $y = 2 + \frac{8}{x}$

$= 6$

P is (2, 6)

(iii) gradient of normal is 1

equation is

$y - 6 = 1(x - 2)$

$y - 6 = x - 2$

$y = x + 4$

c) $y = 3x + 2$ gradient 3

$f(x) = \frac{1}{4}x^2 - 5$

$f'(x) = \frac{1}{2}x$

$f'(x) = 3$

$\frac{1}{2}x = 3$

$x = 6$

$x = 6, y = 4$

at (6, 4) the tangent is parallel to $3x - y + 2 = 0$